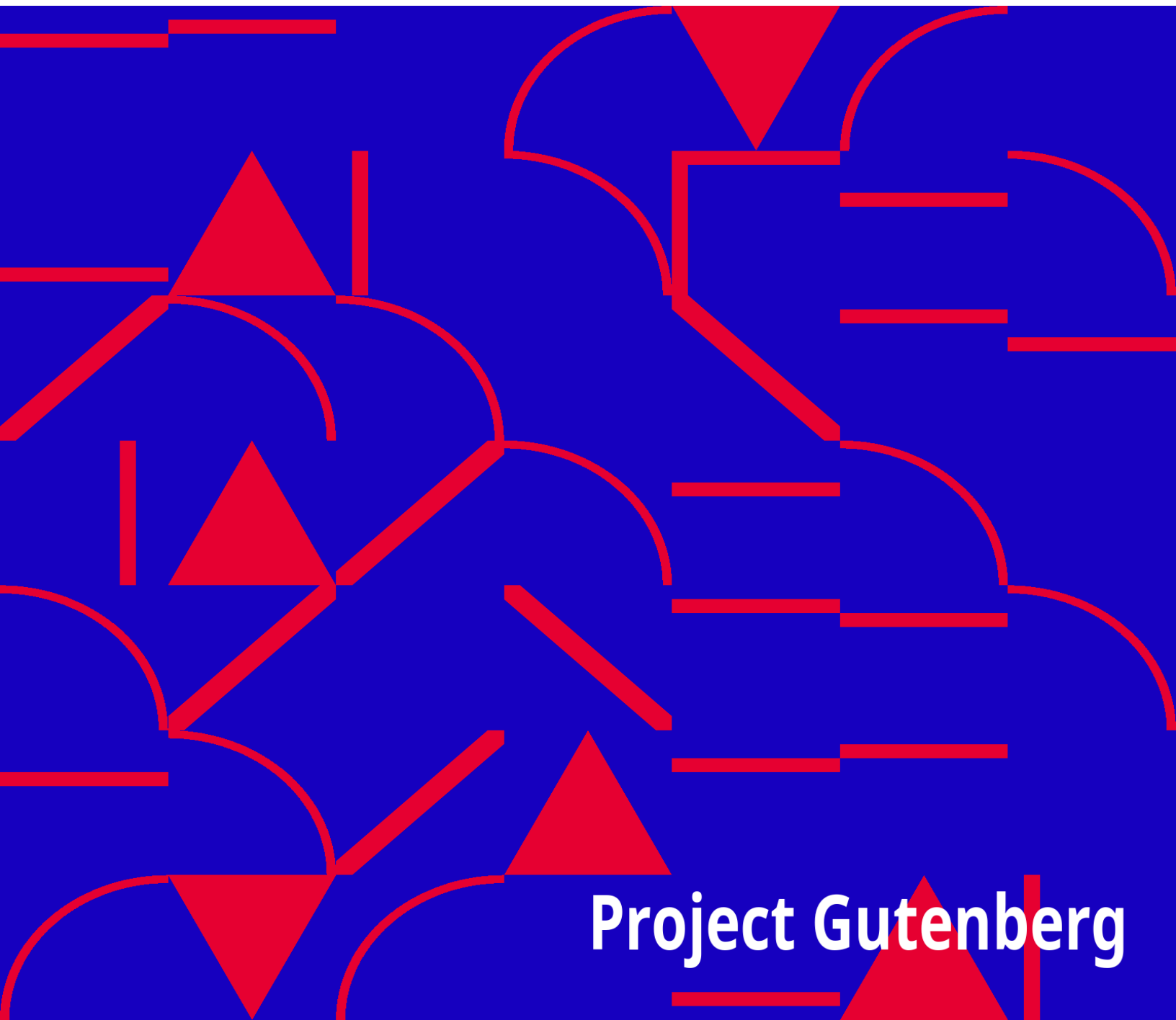


De Usu Ratiocinii Mechanici in Medicina

Herman Boerhaave

An abstract geometric pattern consisting of various red shapes and lines on a blue background. The shapes include triangles, squares, circles, and arcs, some of which are solid red, while others are outlined in red. The lines vary in thickness and orientation, creating a complex, layered visual effect. The pattern is distributed across the lower half of the image, with some elements extending towards the top.

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Herman Boerhaave

The lower half of the image features a complex, abstract geometric pattern. It consists of various red shapes, including triangles, rectangles, and curved lines, arranged in a way that suggests a mechanical or architectural structure. The pattern is set against a solid blue background. The overall design is reminiscent of a book cover or a decorative endpaper.

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Title: De Usu Ratiocinii Mechanici in Medicina

Author: Herman Boerhaave

Release date: April 22, 2005 [eBook #15680]

Most recently updated: December 14, 2020

Language: Latin

Other information and formats: www.gutenberg.org/ebooks/15680

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RATIOCINII MECHANICI IN MEDICINA ***

HERMANNI BOERHAAVE

DE USU RATIOCINII MECHANICI IN
MEDICINA

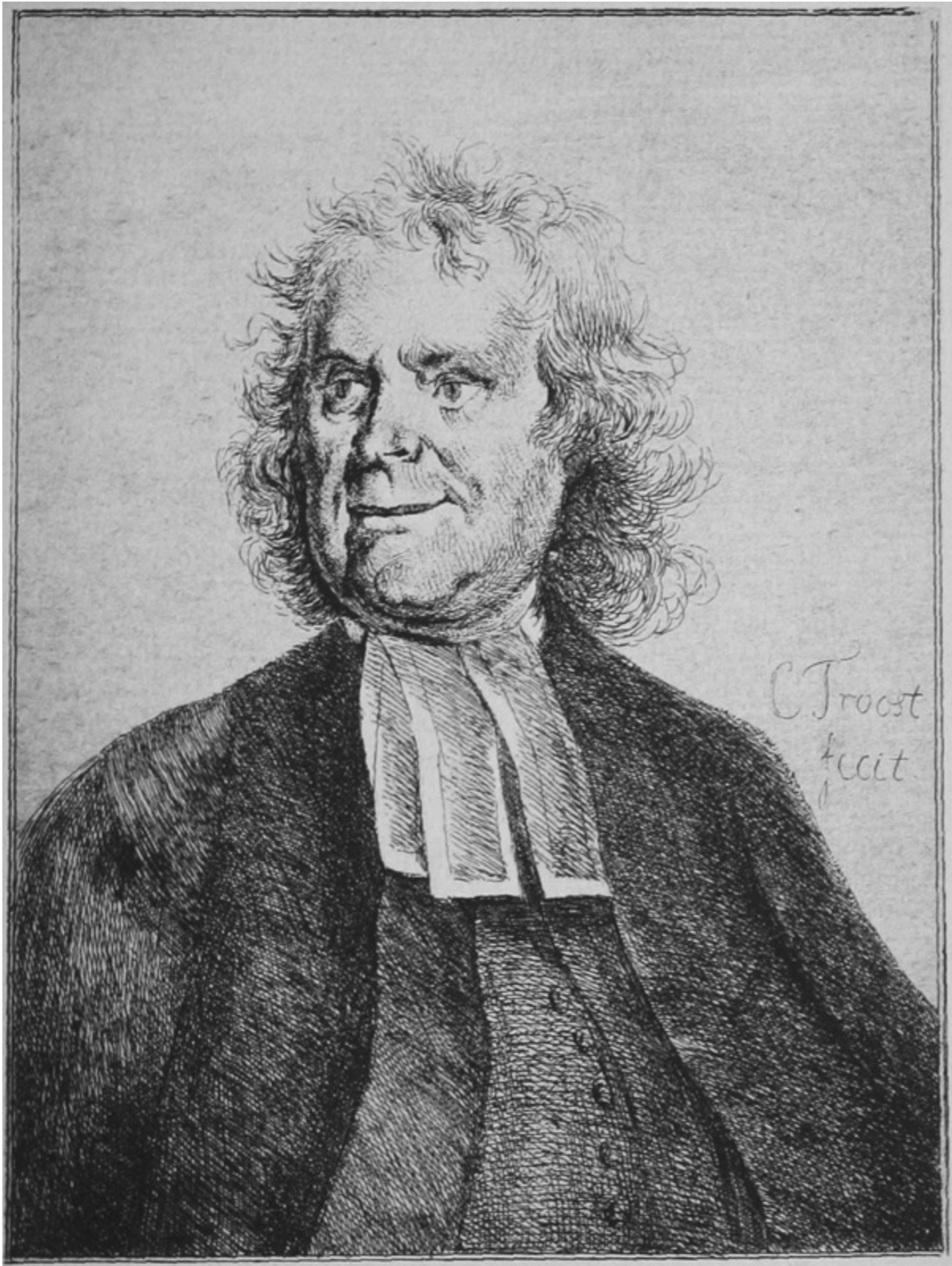
ORATIO

HABITA IN AUDITORIO MAGNO

XXIV. SEPTEMBRIS.

MDCCIII.

CUM TERTII SUAE STATIONIS ANNI LABORES AUSPICARETUR.



Boerhaave.

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HANC ORATIONEM
EA, QUA PAR EST, VENERATIONE
SACRAT
Virtuti, et Nomini Eorum
Devotissimus

HERMANNUS BOERHAAVE.

HERMANNI BOERHAAVE

DE USU RATIOCINII MECHANICI IN MEDICINA

ORATIO.

Qui corporum vires ex mole, figura, et velocitate, vel assumtis, vel deprehensis observatione, calculo aestimant Geometrico, Mechanici appellantur. Quos ipse Artis usus, claraque demonstratae veritatis lux, Sapientibus adeo commendavit, ut aliam omni aequae laudatam seculo, omni aequae comprobata suffragio, temere non inveneris. Miram profecto, et insperato rei eventu humana fere altiore Sapientiam!

Illa enim certis quidem, sed paucis admodum, iisque vulgatis ubique principiis fundamenta debet subtilissimi cujusque et difficillimi inventi.

Postulata ideo Scientiae hujus sordent his, qui fronte prima decepti rebus pretium statuere, vel obscura tantum suspicere solent. Artium vero severissimae successum quisquis spectat, summo eam ingenii cultu dignissimam habet, quia fundamento subnixa tam plano Hominum robur longe supra vires Generis Humani evexit. Ejus quippe effectum nulla datur immobilis moles, licet moturus minimo valuerit agendi momento.

Quare utilitatem ejus omnis civilis, omnis agnoscit militaris disciplina. Hanc aliis artibus necessariam non tantum idonei judices, sed et vanae gloriae ex ignara laude aucupes imperiti celebrant. In sola medicina spernitur, vel praetervisa nihil boni praestare vulgo censetur.

Quod ipsum tamen adeo ego alienum a rei veritate, adeo calamitosum fundo medico habeo, ut dicendi argumentum hac mihi hora aliunde non petiverim. Neque Vestram expectationem, neque mea me vota fefellisse crediderim, si plani sermonis perspicuitate evicero, *Mechanices in Medicina usum esse summum, necessitatem maximam.*

Quae agitati ubertas rei verborum apparatus praecedere videtur. Sed reficit me Vestra in iudicando spectata satis sinceritas, quae damnata dudum exordii demulcentis lenocinia ab loco hoc, qui soli veritati sacer, relegavit. Rem itaque ipsam libere exordior; maxime quum severa veritas patientiam quidem et attentionem impleret, gratiam vero repudiet et odia.

Generalem corporis naturam nullos definivisse verius quam Mathematicos tam clarum habeo, ut litem de fide hujus asserti exspectem plane nullam. Quae vero singulari cuique, prout in rerum natura existit, corpori propria sit indoles, ex universali hac Geometrarum idea a priori nullus rite deduxerit. Illa enim ex sola collectione communium nata, secluso accurate omni eo, quod unum ab alio distinguit, justo ratiocinio non dabit conclusionem unquam, quae peculiarem corporis naturam explicet. Ab hac ipsa tamen pendet primario vis agendi, qua unum prae alio corpus pollet; adeoque illa ignorata et haec incognita lateat necesse est.

Ignota igitur haec detegere quisquis amat, ex ipsa re singulari conditiones eruere debet, quae procacem aliter ratiocinii libertatem in indaganda rei indole exacte determinet. Has vero certo nullus novit, nisi ille, qui sensuum experimento observandos corporis cujusque effectus perspexit. Habent sc. hi rationem eorum, quae ex natura propria rei indagandae fluunt; singula ergo horum unam hujus proprietatem, collecta vero simul integram ejus naturam absolvunt, qua sensibus patet.

Quicumque autem ex his ipsis liquidissime prius perspectis, more dein Geometrico ea demonstrat, quae clara et individua sequela inde elici possunt, plura longe detegit, quam sensuum auxilium revelasset unquam. Neque tamen ipsa haec posteriora vera minus prioribus, neque minus certa, neque minus apta usui erunt.

Praeter binas hasce, tertia non datur, quae peculiarem corporeae cujusdam machinae constructionem reseret, clavis.

Quarum utraque id evincit unum, humanum corpus idem esse natura toti, quam contemplamur, Universitati rerum.

Sensu teste et ratione iudice nil habet praeter caetera eximii, si seria speculatione principia ejus lustraveris, nisi quod ex pluribus, diversisque machinis influxu humorum agitatis illud possidemus conflatum.

Conflatum vero hac conditione, ut adunatarum partium effectus sit plures producere, eosque varios valde, motus, qui mechanica plane evidentia ex mole, figura, firmitate et nexu partium inter se, fluunt. Quod confirmatur satis, quoniam solo mechanico motu destructa harum partium una, vel soluta tantum vinculi tenacitate, frustra eundem deinceps effectum speramus. Humanum ergo verum est, quale Mechanici speculantur, corpus; habet adeoque id omne, quod clara hujus specie exhibetur.

Eadem igitur lege, qua mathematicum illud et humana haec machina explicabilis arti geometricae erit; si modo pro datis assumuntur, non quas arbitrium mentis ex infinita possibilium varietate pro lubidine finxit, sed sensuum usu probe compertae dotes ejus peculiares.

Quarum plurimas anatome vario equidem detexit artificio, observando majorum, quibus componimur, partium definitam structuram. Plura in minoribus pulcherrimum detexit microscopii inventum, similem his, majoribusque naturam demonstrans. Sed et liquidorum scientia revelavit multa, quae humorum per vasa nostra circumactorum ingenium, impetum, directionemque determinant. Quare, aut ex omnibus his nihil lege scientiae deduci poterit unquam, aut soli mechanicae in cognoscendo, adeoque et in gubernando corpore humano palma tribuenda erit.

Nihil veri, nihil certi, nihil quod ex usu sit, ex tot manifestis observatis deduci posse, sive ea quis rite expenderit singula, sive emendatissimo ratiocinio inter se comparaverit universa, quis credet, quis asseret?

Languentis certe animi tardum nimis torporem, et ingratum plane pulcherrimorum, quae possidemus, inventorum neglectum, qui sic loquitur, palam facit.

Desidiosi est nihil agendo desperare semper, vel elevare verbis, facere quae forte solus non possit.

Quod si ratiocinandi lege ignota quidem inde illustrari posse concedens quis, mechanicis tamen solis id muneris denegat, aliam det quaeso, quae corporea rectius excutiat, artem. Id qui aggreditur, necessarium est ut statuatur rerum naturam optime explicari per ea principia, quae a quaesita rei natura maxime aliena sunt, et per eos, qui ab una omni Bono probata veri indagandi methodo longissime aberrant. Eo autem ipso tot, tantisque se

intricat absurdis, ut, nulla ejus ratione habita, propositum demonstratum putem.

Sed jejuna nimis audit haec convincendi ratio, cujusque remotior ab usu communi vis paucos in assensum cogat! Id verum quin sit, si ex plurimorum captu aestimatur demonstrationis pondus, nullus dubito.

Quidni ergo, vel horum gratia, in liquidissima luce locatam rem ponamus ob oculos; et in ea quidem, qua se omnes pulchre uti jactant, quibus mederi cura est.

Quae aggressurus vel invitus sane cogor ex historia structurae corporis allegare ea, quae Rhetorum locis insueta plane et inaudita, puritati defaecatae Latinitatis peregrina et barbara, intellectui tamen ipsius rei praeprimis necessaria habentur.

Maximam corporis nostri partem arteriis contextam, harumque sustentatam beneficio vigere, clarius est, quam demonstratione ut egeat. Has canales esse cruorem qui castigant, inque suo dirigunt itinere, quorum maxima circa cor sensim gracilescit cavitas, donec prae tenuitate aciem visus fugiat, vel laniones norunt. Neque minus vulgatum, a corde exortum unum horum truncum explicari in ramos laterales, figura trunci similes, eadem ratione et divisos rursus et decrescentes, hoc tamen artificio, ut truncus recta pergens, in loco divisionis majori plerunque capacitate aperiatur quam rami, qui ad latera trivii hujus porriguntur. Sinuoso autem flexu ita haec omnia vasa curvari, ut cavitatum latera ad infinitos numero, et magnos valde angulos ubique inflectantur, hujusque Spirae gravissimos effectus esse in sanguinem transfluentem, observarunt a paucis retro annis, qui Geometricas subtilitates rebus applicuere Medicis.

Quam mirabili vero, quam efficaci fabrica flexiles finxit hos canales Adorandus nostrae machinae Faber!

Dum a premente intus liquido distendi posse sine lacerationis discrimine voluit, eoque rursum fecit ingenio, ut humorem a dilatatione reciproca cessantem valido cum impetu cogere, se vero in arctiorem capacitatem propria sponte restituere queant.

Ultimos autem arteriae, hosque minutatim divisos fines in membrana, ut firma basi, ordinari, ibique per fistulas in mutuos occursus emissas hiare

inter se, ante Malpigium viderat nemo. Ille primus ambages resolvit et mille viarum dolos, quos pulsa in hos Maeandros liquida pererrant.

Sed, o admirabilitatem maximam! o mechanismum pollicis divini!

Tanta enim accuratione digesti ramuli aequali hic viae latitudine porrecti et laterali progenie orbi, primordia venarum, Lymphaeductum, horumque sinus mutata constituunt figura.

Haec ea sunt, quae oculi acies, microscopium, vasorum in vivis ligaturae, hydrargyrium mortuis injectum, contemplatio figurae morbosae, comparatio denique brutorum, piscium, insectorum et plantarum detexit.

Praeter illa in arteriis ipsis deprehenditur nihil, falso finguntur plurima.

Maxima ergo corporis, eaque efficax valde ad vitam pars, Mechanica descriptione, canalis est conicus, elasticus, inflexus, divisus in similes minores eodem trunco ortos, qui ultimo circa vertices cylindricos retis structura in se mutuo patent.

Id si verum, quod omnium profecto verissimum, nonne sequitur omnes effectus quos sanguini arteriae praestant, tantum pendere ab hac earum fabrica?

Nonne et hoc rursum liquet, omnes ergo illos hinc solummodo petendos, et demonstrandos esse?

Vos nunc, qui justi sedetis hac in causa Iudices, obtestor! Quis ea, quae vel hinc duntaxat oriuntur, verae demonstrationis ordine expediet?

Solus ille, qui figurarum contemplationi, et oscillatoriae virtutis calculo assuetus, callide videt, quam multa, quam gravia ex hisce solis demonstrare queat; solus ergo Mechanicus.

Sed patiamur abripi nos admirabilitate hujus arteriae, brevis certe levisque attentionis praemium Scientia erit totius fere humani corporis.

Illa, ubi depictum antea rete constituit, tubos emittit cylindricos adeo arctos, qui rubras cruoris sphaeras ore suo capere nequeant; unde his recipitur tenuior tantum et excolor pars sanguinis.

En veram vasis lymphatici ideam!

Eadem rursum ibidem loci arteria recto porrigit decursu truncum, qui emissis Lymphaticis amplior crassiorem, rubrumque sanguinem, sero liquidiori orbatum vehat.

Ecce venarum genuinam originem!

Quarum angustam primo cavitatem mox ampliorem reddit infusa ubique nova per laterales fistulas liquidi venosi, Lymphaticique moles, prorsus ut novum conum, similem arterioso, eique ad vertices oppositum repraesentare discat.

Perfunctorie tangere quae debui, vasa, vah quae, quamque pulchra in recessu recondunt!

Arterias, Venas, Lymphaeductus, descriptumque horum apparatus plano affigas membranaceo, huic nervos intexas, villosque applies elasticos, tum convolvas in glomerem, habebis glandulae fabricam.

Quam quoties cogito, uberrimam mirandorum effectuum matrem contemplor, simulque ineptissimi cujusque figmenti falso celebratam sedem.

Tu vero inanes Chimaerae latebras aperiens, Tu maxime Malpigi! Suprahumana industria, incredibili labore, atque cautissima perspicientia, simplici hoc artificio absolvi ejus compagem, plus quam demonstras!

Quanti vero momenti demonstratio! glandularum enim aggregato totum fere corpus constat!

Cerebrum Hippocratico oraculo glandula penicillo Malpigliano depingitur ut ordinata ex arteriis, venis, receptaculis, emissariisque nervosis moles. Jecur, Lien, Renes glandulis fiunt adunatis.

Ipsa humoris genitalis officina artificiosus canalium cylindricorum glomus. Ipsum Embryi dolium, ipsa foetus aula, ipse candidi nectaris, quod recens nati bibunt, promus condus hac glandulosa operantur arte. Ossa ipsa et membranas eadem fere compaginari structura quis dubitat, nisi cui cedro digna et aere scripta Malpighii, Kerkringii, Havertiique nondum illuxere?

Lacertis tandem examinandis mentem applicuisse rogo ne poeniteat! Huic se labori quicunque non subduxerit, nae ille subtilissimae Mechanicae artis efficacissima instrumenta clarissime reperiet! Musculus enim omnis nonne ex minoribus similibus componitur? Ultimius vero quid, quaeso, villus est? Non aliud certe, quam nervosi et angustissimi canalis dilatata, simulque attenuata pellis canali, unde oritur, cavum formans amplius soloque inflatum spiritu.

Hujus vero quam immensa sit machinae potentia, scite novit, qui hydraulica Mariotti experimenta contulit Cartesii Mechanicis.

Pulmones contempleri, diversae a caeteris structurae, saccos habebitis elasticos, sphaeroïdeos, qui abscisso coni vocalis appenduntur vertici; horum superficies maculis retis sanguiferi ornatur, et, quod mira hic arcana velat, incilibus fere caret lymphaticis.

Ergone, cogitatis forte, admirabilis illa, illa tam artificiosa Hominis machina simplici adeo perficitur apparatu!

Certe non fit alio.

Habeat hanc, qui volet, ob simplicitatem, vilem!

Mechanice Organum id laudat, ejusque Auctorem celebrat sapientissimum, quod quaesito effectui producendo aptissimum, simulque inter omnia, quae eundem praestare possent, simplicissimum sit.

Quid tandem ex hisce concludemus?

Corpus nempe humanum machinam esse, cujus solidae partes aliae sint vasa liquidis coërcendis, dirigendis, mutandis, separandis, colligendis, et excernendis apta; aliae vero instrumenta mechanica, quae figura, duritie nexuque suo vel fulcire alia, vel definitos motus exercere queant.

Peccabo in patientiam vestram vestrumque decus, si cuncta examussim explico. Id unum bona audietis cum gratia: Hippocratem cum integro, quem sequutus est Babyloniorum, Ægyptiorum, Graecorumque choro, cum integra, quae eum sectata est Grajorum schola duo haec, non alia detexisse.

Arabas omni industria, omni anatomes cultu tertium addere potuisse nunquam.

Instauratorem anatomes consulite Vesalium, hujus aemulos Eustachium et Fallopium; tum immortales inventis Harvaeum et Malpighium; et hos, qui singuli novis antiqua emendarunt Asellium, Pecquetum, Bartholinum, Dathirium, Bellinum, Glissonium, Wharthonum et Willisium; his jungite juxta leges mechanicas anatomicos Lealem et Louwerum, quique in abditissima penetrarunt, Hokium, Pouwerum, Leeuwenhoekium, deprehensuri estis omni arte, omni artis adjumento bina, quae dixi, nec inventa alia.

Cur alia ergo fingere precario quempiam patiemur, nobisque imponentem in aeternum verba dare?

Ubi Elementis, qualitatibus, formis, causis chemicis, animatis, metaphysicis, amoris et odii affectibus, ubi, inquam, tot fabulis locus, causa, necessitas?

Nulla profecto vel vestigium sui hic figmenti secta invenit.

Soli Mechanici suum objectum hic agnoscunt, neque aliud in toto, qua solidum est, corpore quidquam datur. Ille ergo soli audiendi, horum effata sola consulenda, eorum principia sola imploranda, horum methodus sola adhibenda, ubi de effectu organi perspecti quaeritur.

Sola erit firma, quae a perito in his Magistro profertur, demonstratio.

Agite o Viri, queis dicta forte displicent, quid facit in oculo vel simplex illa figura corneae, quid aquae, quid crystallinae lentis, quid vitrei humoris determinata superficies et definita spissitudo?

Enarrate quid auris externae Helices, quid meatus auditorii arctior et inflexa in medio, latior et porrecta ad utrumque extremum via faciat ad exceptionem, directionemque radii sonori?

Membranae Tympani tenuitatem, figuram ejus ellipticam versus interiora ossis petrae convexam, hujus mutabilem in varias curvaturae figuras formam ope affixi et agitati suo musculo malleoli contemplerini, et dicatis, quis effectus constantissimae hujus tamque operosae in vilissimo quoque animalium fabricae?

Nunc daedalei labyrinthi, conchæ, vestibuli, duplicis in cochlea turbinata spirae, loci ovalis et rotundæ fenestræ, tot inquam miraculorum mechanicorum, quae durissimae hic insculpsit petrae Divina manus, date rationem.

Sine profunda Mechanices Scientia nil veri vos intellecturos, nil boni proluturos aliis, utamini quolibet adminiculo, audacter affirmo.

De solidis, quae dixi, pauca haec sufficiant; urget ratio ut nonnulla de fluidis subnectam.

Haec enim illa sunt, quorum motu vita, quorum libero per vasa fluxu sanitas absolvitur.

Illorum autem naturam exacte capit, qui minuta novit corpuscula et agitata, quorum congeries fluidum constituit. Eorum unum si spectatur, rationem habet solidi, adeoque mole, motu, figuraque quidquid agit, efficit. Quare effectus, quos una fluidi pars producit, soli Mechanico patent per experimenta indagandi.

Quod ex ante dictis quum sponte fluat sua, latiori sermone non explano; unum hoc pronuncians, non eo usque hactenus provectam hanc liquidorum scientiam, quae usum rei praestet idoneum.

At si totam fluidi molem simul spectamus, gravitas ejus fluorque communes deprehunduntur sublunaris liquidi proprietates. Virtus vero elastica, ponderis, spissitudinis, fluiditatis, nixusque in contactum gradus varii, momentum impetus quo fertur, et itineris directio palmaria sunt quae unum ab alio fluidum distinguunt. Horum vero omnium tanta efficacia est, ut infinita, quae sanis contingunt, non aliunde oriantur.

Quamobrem quicumque ex praecepto scientiae rite haec enucleat, opus is absolvit summae ad perfectionem medicam necessitatis.

Sed fidem vestram! quis proponere, explicare et demonstrare vim eorum poterit, qui Hygrostatices, quae subtilis Mechanices pars, rudis est?

Haec illa est Aquilegum scientia, quae ex assumtis, modo quas descripsi, affectionibus ratiocinia nectens geometrica utilissima et usui apta reperit Theoremata.

Haec, neglecta causa physica, et cujusque particulae, quae fluit, singulari natura, ex his, quae sensibus per eventum in tota mole patent, quam gravia, quam utilia vitae, methodo invenit Mathematica?

Evolvatur Archimedis, Cartesii, Stevini, Borelli, Mariotti, Hugonii, Neutoni, et Bellini scripta, qui re, non verbis, convinci cupit.

O quam necessaria feliciori Genio, ut revelentur, reliqua sunt in Pulcherrima hac Speculatione!

Hanc utinam excolant! utinam exhauriant! utinam nobis aperiant Viri Mathematicae docti!

Ab hoc Eorum labore, quo generales liquidi effectus luce illustrarent mathematica, brevi tempore plus maturi in horto medico fructus exspectare licet, quam ab omni eo, quod aliunde in hunc congestum hactenus.

Taedet quippe pudetque ineptiarum, quibus seriam prae caeteris Artem ridiculam fecere, qui Mechanices imperiti vim liquidorum humanorum explicare conati sunt.

Et palam affirmo, vitalium actiones humorum scire posse neminem, qui Aquilegum regulas ignorat.

Quae dum libertate Medica firmus assero, jurgii hic illaturos causam praesagit animus eos, Qui, nescio qua gratia, ab Hermete nomen sibi, sectamque condunt.

Egone ex universali hac liquidorum doctrina deduxerim ea, quae singulares eorum virtutes absolvunt?

An fermenti stabiles motus, diversorum liquidorum ferventes conflictus, putredinis spontaneae mirabiles effectus ex Mechanicis explicuerim unquam?

Talia objectans, eorum, quae dicta, memor, paucis, quae dicam, animum adhibeat.

Mea enimvero sic est ratio, justa, vel secus, vestrum sit judicium.

Ex experimentis Chemicorum historiam haberi posse valde limitatam singularium eventorum, quatenus in circumstantia definita sensibile quidpiam producant.

Necessaria ergo quam maxime est Medicinae haec Ars, dum observatorum Sylvam largitur et observandi praebet optimum compendium.

Data enim exhibere, horumque definire conditiones valet, regulas autem ratiocinandi ex his Chemia dabit nunquam.

Ne tamen vel sic nimis, ut solent, se efferant, qui unius Chemiae cultu omnem Medicae Sapientiae thesaurum se possidere vani jactant!

Enimvero plura in nobis, sani vigeamus, vel langueamus aegri, fieri ex communibus illis liquorum proprietatibus, quas sibi sumserunt expendendas Geometrae, quam ex insitivis, dubiis, et arte Chemicorum factis plerumque, pervulgato palam documento est.

Aqua naturae ariditatem alter corrigit, Falerno alter quotidie venas inflat; fructibus hic, Cerealibusque parvo assuetus famem explet, et sustentat Spiritum, ille carnibus, piscibus, terra natis, et omni condimentorum varietate Apitiana onerat ventrem; alii blando et insulso fere victu aluntur, alii salitis, acidis, et acris quibusque intestina stimulant.

Multiplex adeo assumptorum varietas vitam tamen sanitatemque plures per annos protrahit in iis, qui tamen diversis humores suos saturant corpusculis.

Liquido argumento magis communi fluidorum naturae Mechanicis explicatae, et in ipso corpore vi viscerum productae, quam singulari cujusque particulae virtuti, actiones vitae deberi.

Si aurea Verulamii de vita et morte monumenta, si liberae Hippocratis et Celsi de victu sanorum leges, si usus non satis id confirmat quotidianus, omni dignissimum fide Louwerum, sincerum mehercle et defaecato iudicio sagacem Virum vobis citabo.

Hic enim, immani cruoris jactura exsanguem, jure carni solo ingesto, venis recepto, per has fluente, imo colore nec mutato effluente per vulnera, revixisse Juvenem testatur.

Sed quid verbis opus in re clara?

Ad Vos ego provoco, Vestram appello fidem Clarissimi Viri Medici, Quorum sapientia huic Coronae venustatem conciliat, Quorum salutari dextra incolumis huic Urbi praestatur sanitas!

Nonne incumbit nobis, dum aegris Medicina fit, vel millies fluida inspissare, resolvere coacta, stagnantia movere, compescere dissoluta, diluere crassa, leviora solidare?

Dum rarissime ad pugnas Salium, flammas Sulphurum, vel tectum Mercurii genium attendere cogimur.

Ipsi certe illi, qui mera ubique Chemica crepant, cum morbus manum poscit, repudiatis suis, sedulo, quae laudavi, inquirunt.

Si ergo his fluidorum proprietatibus tot debentur, si has omnium suffragio optime excusserint Mechanici, patet ipsa fluida vitalia ut cognoscantur Medico, auxiliis egere Mechanices.

Spectate jam effectus, qui ex fluentibus per vasa liquoribus oriuntur, evidentior longe fulgebit Veritatis Mechanicae potestas.

Si enim liquida descripta in vasis depictis quiescunt habebimus cadaver.

Ubi vero liber his humoribus per canales conciliatur motus corpus vivum cernimus.

Sermoni fidem quisquis meo negat, suis ut oculis credat oportet.

Mollem consideremus hominem, qui salientis de vulnere cruoris spectaculo perturbatus in animi cecidit deliquium.

Mortuum videmus; sed qualem? in quo cuncta solida, quae sanitati sufficiunt, adsunt et liquida, solus abest liquores in gyrum agens motus.

Huic quacunque demum ope concutiantur nervi, ut motrix cordis materies fluat, redit statim, depulsa tristi mortis imagine, laetior vita.

Vita non modo; calor, rubor, agilitas, cogitatio, vitalis omnis, naturalis et humana simul redit actio.

Quid hic fermenti, quid effervescentis, quid salis pugnacis, quid olei spiritusve nascitur aut perit?

Excepto motu, neque additur, neque demitur quidquam, vita tamen amissa ipsa redditur.

Sic aves et insecta constricta frigore hyberno, lenis statim in vitam excitat tepor.

Sed veritatis qui convictus viribus, ob ipsam argumenti vulgatam claritatem, certis saepe diffidit.

Rariori ergo ut spectaculo firmetur, quae nimis noto patuit satis exemplo fides, in Hokii vos officinam invitat oratio.

Destructo thorace mortuum animal inflatis per follem Laryngi applicatum pulmonibus cito reviviscit.

Attoniti miraculo vitae tam mechanicae ad magnum cito adeamus Glissonium; en ille impulso ope vesicae in venas liquido mirifice vitales actiones aemulatur in defuncti dudum hominis cadavere.

Omnia haec in specimen allata, infinita enim dici possent, an non evincunt satis, cuncta fere, quae vitam, sanitatemque nostram faciunt, vel sequuntur, pendere a motu illo, quo humores per vasa mutua plane moventur et agunt vicissim agitatione?

Cujus effectus, et leges, quum soli rite intelligant, explicant, et demonstrent, in Pneumaticis atque Hydraulicis, Mechanici, concludo cuncta ergo rursum disciplinae subjecta haec Mechanicae.

Hic vero ille est locus, ubi mire se jactant, ubi serio triumphant fermentorum Patroni.

Si fluor liquorum liber per vasa vitae causa, ergo ajunt prima motus ratio in fluido et ab eo; itaque ab interna huic agitatione, eaque forti valde et constanti satis, qualis non nisi in excitatis fermento liquidis reperiunda datur.

Sciant autem Hi, primam moti in Embryo liquidi a parentibus semper derivandam causam, eam motu matris continuari dum ab ea pendet foetus, dein vero ab ipsa fabrica perennare solidorum.

Admirabilem auricularum Cordis ad ejus Thalamos structuram, nexumque qui speculatus est, et qui hinc necessario sequuntur, alternos influentis et expulsi liquoris motus a corde in arterias, ab his in cerebri medullam, processus, nervos, musculosque et venas rursum, non quaeret vitae continuatae rationem extra ipsam virtutem viscerum Mechanicam.

Facile enim illi erit, perspicuitate certe Mathematica demonstrare, unicum pulsum cordis datum in corpore sano sibi continuando esse causam.

Longe minora numero, longe simpliciora sunt, quae vitae incolumitatem praestant, quam noster fingit animus.

Leviores longe sunt rerum ingestarum in nobis mutationes, quam vulgo creditur.

Minus compositae, quam ipsi putamus, vitae humanae causae.

Si exacta structurae esset cognitio, si sensibilis probe nota esset humorum natura, doceret cito Mechanice ex simplicissimis fluere principiis, quae ignota maximam nunc pariunt admirationem.

Dicti veritatem tam paradoxo uno ab exemplo discere licebit, ut constet quam simplici negotio et Mechanico plane maximae quae habetur omnium operae mutatio in nobis fiat.

Pars pellucida animalis vivi microscopio aucta claro docet spectaculo, cruorem solo cordis pulsu in extremas trudi arterias, ibi elastica arteriae contractione retropelli aliquantulum quo momento ictus cordis cessans, ejusque valvulae concidententes, regressui spatium laxant.

Reciproco hoc impulsu et percussu varias mole partes cruoris applicari ubique ad diversa capacitatis hiatus oscula, intra haec recipi, vel inde repelli, tam clare, quam coelum hoc contueri est.

Tum solo hoc artificio secedere sanguinem in diversa colore et tenuitate fluida, mox in venis iterum permiscenda eadem claritate cernitur.

Id vero Chemicorum conflictuum perito evidens ipsi oculi aciei apparet, simplici impulsu aliunde dato, et vasis elatere, sine ullo fermenti signo omnia haec fieri.

Defixus saepenumero in speculatione hac anceps mihi haesit animus, an Spirantis cerneret animalis partem, an vero incilia meditatione summi Mathematici excogitata, manu peritissimi Mechanici affabrefacta, per quae liquores duceret, secerneret, misceretque absolutae artis consummatione perfectus Aquilex.

Tandem vero si periculum capere juvat, an ex simplicibus et indubitatis sensuum experimentis demonstrari queant per Mechanicos illa, de quorum intellectu ante paucos annos nulla spes, Geometrico parta labore in usum exempli citare decet.

Perpendamus, quae docet, dum Mechanicen Medicis applicat Rebus, Borellus.

Evolvantur, quae ex hujus Schola sapiens, eisdem usus principiis, et Malpighianis inventis fretus Oedipi instar extricat Bellinus.

Tum quae illorum laudato excitatus labore, Orbi erudito Problemata proposuit, demonstravitque, nobile quondam hujus Lycaeii ornamentum Pitcaernius.

Scheineri, Cartesii, Hugonii de oculo, Kircheri, Schellhammeri, et Morlandi de aure et auditu, scrutemur demonstrata.

Constabit an prosit Medico Mechanice!

Apparebit quid sperandum sit, si ejus a peritis Medicis invehitur in Medicinam usus, si in exercitatione hac pergitur tamdiu, quamdiu patientia humana tam inepta sectarum molimina in disciplina Medica tulit.

Haec autem vera esse, et usum habere in Medicis Mechanicen, quamdiu de Theoria agitur, consensus erit forte facilis, tamen ne hilum bonae frugis ipsi Artis exercitio afferre, pervolgata objicitur querela.

Quae quidem speciosa hac distinctione prolata, qui consistere queant simul, satis non video.

Neque enim aliam hos intelligere Theoriam credo, nisi eam, quae ex proximis causis clare docet, quae sani hominis vita sit.

Quod si, ut oportet, admittitur, sequetur Scientiam hanc noscendis, curandisque morbis auxilia suppeditare optima.

Causas enim qui recte novit perfectae sanitatis, ille, quoties hae deficiunt, egregie ipsius defectus, id est morbi, originem rationemque comprehendet.

Qui autem causam aegritudinis proximam clarissime vidit, maxime is idoneus, qui ei occurrat, est habendus.

Eodem sc. modo se res habet ac in horologio, cujus si deviat index, errores imperitus notare, at corrigere ex arte nemo potest, nisi ille, qui requisitae structurae gnarus, vitia partium hinc et remedia invenit.

Ita nulla lucis scintilla in Theoria Medica micat, ad quam in faciunda Medicina facem accendere non possit re peritus Artifex.

Adeoque qui Mechanics in Speculatione, ille ejus in usu praestantiam fatetur.

Docet hoc antiquitate nobilissima et usu ea artis pars, quae ab eo quod manu medetur nomen gerit, quae sc. an inventis Mechanicis carere queat vestra sit aestimatio.

Instrumenta, quibus vitia emendat, quis felicius, quam Mechanicis imbutus Medicus inveniet?

Tenues, quae volitare putantur ante oculum, imagines, dum Matheseos imperiti ut oriturae in aqueo humore suffusionis primordia tractant, acerbis saepe erodunt tenellum et prava arte oculum.

Harum vero sedem reticulo, causam arteriis Geometrae consilio dum reddit Willisius, dum demonstrat Pitcarnius, quam mutata est medellae facies?

Abacto externorum mordaci apparatu, misso sanguine, et solventi medicamine tuto tollitur, vel et negligitur malum.

Oculi error a radiis male collectis quam inepte tentatur collyriis vel potus medicati haustu!

Quam feliciter levatur perspicillis, quae cuique vitio singulari propria regulae definiunt Hugenianae!

Opto ut, qui omnem Mechanices usum ex praxi proscribunt Medica, intelligent prius vel unius Hugonii de emendandis visus vitiis Commentarios.

Illustre enim illud Batavorum lumen, assumpta ex anatomicis oculi fabrica, et una morbi, cui succurrere vult, proprietate, mox ex meris Mathematicis reperit auxilium, quod usum praestat huic tantum malo, cujus proprietates assumpta problema limitaverat.

Intacto oculo, morbi effectum tollit; et inemendabilem in eo defectum vitri figurati supplemento farcit.

En pulchra, in quibus, ut in speculo, spectatur Geometrarum in medicis Mechanice ratiocinandi methodus, usus et successus.

Hac via si pertractabunt omnia, ut revera sensim poterunt, habebitur tandem certior, neque obnoxia figmentis, neque omni mutabilis hora, sed aeterna scientia medica.

Non est porro quod dicat quis, nondum confirmari vitia fluidorum adeoque internae aegritudinis causam, hujusque mitigationem auxiliis subijci Mechanicis.

Vel enim an impossibilis fructus hic, vel an necdum acquisitus quaeritur.

Si posterius, iniquos habemus et molestos Censores.

Quis aequo ferat animo peti, ut pauci Mechanici, qui Medicis a paucis temporibus vacarunt rebus, ea jam perfecerint, quae tribus annorum millibus junctis viribus alii omnes vix potuerunt inchoare?

Imo id omnino impossibile: quum enim Mechanices Medicis applicandae lex exigat, ut structura solidorum, natura liquidorum, effectus horum sensibiles in sanitate et morbis inserviant pro datis, quis tam absurdus, qui operosissimae Artis fastigium in ejus rudimentis quaerat.

Si autem judicat quis nunquam vel quidquam hac via perfectum iri, is, rogo, perpendat, morbi a fluido orti causam pendere *ut plurimum* a vitio ejus per vasa transfluxu.

Hoc Hippocratica, si componuntur Sanctorianis et quotidiani usus experimentis, docent.

Fluxus vero impedimentum internum vel languori virtutis impellentis, vel contractioni vasculorum convulsivae, vel liquidis copia, motu, spissitate, aut tenuitate peccantibus adscribet *plerunque*, qui vitae, sanitatis, morbi, mortis et cadaverum phaenomena comparavit sedulus.

Quin adjumenta, quibus morborum miseriam lenimus aegris, ea prodesse gratia *inprimis*, qua dicta malorum capita auferunt, attenta nos docet contemplatio.

Aurea comparentur Sydenhami observata demonstratis de missione sanguinis, stimulis et Villo contractili Bellinianis, et, postquam Mechanica plane ope juvare vulgata remedia constat, spes concipietur sensim demonstrandi regulas subire posse et vires eorum et applicandi rationem.

Vix enim me contineo, quin, praematurius forte, pronunciem simpliciores esse, et magis Mechanicas morborum maxime compositorum causas, quam ullus Medicorum cogitat.

Unius enim partis minima et simplicissima labes unionis necessitate et contagio totam saluberrimae Machinae vim subito pervertit.

Tenuissima acu, eaque ex purissimo Chalybe pungatur tendinis vel nervuli fibrilla in corpore sanissimo.

Heu quam dira ex vili vulnuscule tantillae particulae malorum, heu quam multiplex cohors!

Dolor, rubor, tumor, ardor, pulsatio, febris, sitis, delirium, convulsio et horrenda tristis tragoediae catastrophe mors.

Spina, levisve festuca membranoso infixae loco eadem brevi parit.

Et miramur venenorum spicula, pestis lanceolas, vel salium acumina similia peragere?

Quin solo motu externo quam mirae rerum mutationes in corpore sano!

In gyrum agatur, vel jactetur maris fluctibus scaphae insidens insuetus: Quid fit? vertigo, pallor, nausea, vomitus, anxietas, mille morborum aerumnae, mille fluidi vitalis et incredibiles mutationes a solo motu oriundae.

Qui ergo humores integros manere novit, quamdiu vi canalium conquassati propelluntur, qui stagnantes hos in calido, humidoque loco morbosos reddi statim et trahere sincera scit, qui ex uno simplicique malo infinita alia statim sequi animadvertit, facillime perspiciet exspectanda ad haec a mechanico medico promptissima tandem auxilia: ex causis enim impediti fluoris, regulis superandae resistentiae, restituendi motus elastici, augendae virtutis cordis collatis cum morbi phaenomenis quid non invenietur tandem?

At enim vitam, morbos, sanitatem in nobis ex principiis fluere non Mechanicis mentis docet in corpora potestas. Frustraneus ergo tot irritorum conaminum labor! Vana supervacaneae Mechanicae speculationis spes.

Talia aggerens utinam rideret securus, neque communem ignorantiae calamitatem eadem deploraret querela!

Quis enim miri hujus commercii vim invenire potuit in aliquo, quod corpus constituit vel mentem?

Sciat tamen, virtutem cogitationis, simulac in corpus influit, totum quod in eo producit, facere corporeum, adeoque legi Mechanicae obediens.

Quid refert causam mutationis primam non esse Mechanicam, quum hac insuper habita, effectum, qui corporeus, cognoscere, excutere, atque dirigere Mechanico detur Medico; quum hoc scopo sufficiat?

Crescit nimium, pauca dum tangit leviter, Oratio.

Unum, quod palmarium jactant, quibus alia quam nobis mens est, ne declinando subdole evitasse me suspicentur, diluendum judico.

Philosophos clamant et Mechanicos, ubi Medicae arti exercendae admoti fuere unquam, sinistro semper eventu repulsos fuisse. Disputatione non esse opus, quum artem horum Medicis nocere, re constet et experimento.

Quae verissima esse, si hos arguunt, quos in scholis superbus philosophi titulus effert, docet historia, docent, quae de rebus conscribere medicis,

volumina.

Dum enim omnium prima rerum principia ex propriis creare cogitatis satagunt, dein vero ex iis, quae ipsi figmenti subtilitate prius in illis posuerant, peculiarem corporis cuiusque naturam declarare, errasse ubique docet ipsa, quam commendo, Mechanices ratio.

Applicari rebus nequit, quam ratiocinio fecerant, conclusio, nisi prius illa, quae pro fonte argumenti liquido assumserant, rerum singularium, quae natae sunt, principiis esse eadem foret evictum.

Haec vero, quum infinita, eaque semper diversa esse queant, patet casu veritatem nunquam sic detectum iri.

Quod si considerassent sedulo, tam Scholastici dicti, quam plurimi Mechanicorum Cartesii sequaces non fuissent arbitrati id sibi datum negotii, ut ex fictorum principiorum praeceptis corpus humanum regerent, sed ut ex his, quae observatio prius docuerat hominem constituere, ipsa dein artis elementa applicata Mechanica conderent.

At si Mechanico, quem jam descripsi, Medico hanc dicunt contumeliam, exempla ignominiae citent exspecto.

Non equidem, qui nostri capit animi sensum, negabit ullus, accuratissimum Mathematicum pessimum forte futurum Medicum.

Quo enim talis pertinet Oratio?

Non in Mechanico Medicinae, in Medico vero Mechanices peritiam desidero.

Usu peritum Medicum experimentis medicis defecto Mechanico in morbis curandis qui post habet, insaniet.

Sed aequa instructorum experientia hunc promovendae arti meliorem, qui Mechanicis callet prae alio praeceptis, id affirmo, id demonstrandum sumserat Oratio.

Ne vero, quod ubique contigisse doleo, sinistram, quae dixi, interpretationem subeant, age describam compendio speciem illius, cujus imago animo obversatur meo, Medici.

Depingitur ille, ducendis studii Medici primis lineamentis incumbens, tanquam affixus Geometricae contemplationi figurarum, Corporum, Ponderum, Velocitatis, Fabricae Machinarum, et, quae inde oriuntur in alia corpora, Virium.

His dum mentem exercet, claro discit praecepto et exemplo, liquida ab obscuris, a falsis vera discernere, et ipsa iudicandi tarditate animo conciliare prudentiam.

Ita postquam nudas simplicium corporum actiones expendere, has ex veris, clarisque causis deducere novit, maturum habet ingenium, qui fluididatis, Elateris, tenuitatis, ponderis, tenacitatisque in fluentibus proprietates ab Hydrostaticis cognoscit.

Jam animi vigore robustior fluidorum vires in machinas, harumque in illa rigore addiscat Mathematico, Experimentis confirmet Hydraulicis, et Mechanicis, Chemicis illustret, Ignis, Aquae, Aëris, Salium, et aliorum maxime similium corporum ingenium speculatus et actiones.

Altera mox tabulae facies sacris jam Medicis admotum exhibet.

Oculum ibi Geometriae luce acutum ad incisa cadavera, ad spirantium corpora brutorum aperta tacitus circumfert.

Jam vasorum structuram, figuras, firmitatem, ortum, fines, nexus, curvaturas, flexilitatem contemplatur et elaterem.

Excitatus spectaculi mirabilitate, mox conspecta ad eum, quo jam pollet cognito, Mechanismum applicans, abditas detegit harum partium virtutes.

Quam variis, pulchris, utilibusque utentem cernimus auxiliis, quibus recentiorum industria pmoeria extendit anatomes.

Aliorum certe durissimo parte labore inventa in suos usus dum accommodat, claram sibi sistit humanae fabricae imaginem.

Cui fluidorum vitalium nectit notitiam; hanc Anatomicis, Chemicis, Hydrostaticis, ipsiusque microscopii adjumentis in vivo corpore, et extra illud examinat; tum mox accuratissimam omnium sensibilibus, quae in sanitate contingunt, historiam omni arte, undique comparatam evolvit.

En suis instructum datis, ut sanitatis Theoriam scribat!

Ex his singulatim perspectis, expensis, comparatisque inter se, auxilio
Mechanices, severitate ordine et prudentia Geometrica, lento gradu
festinans elicit, quae in his comprehensa sensibus abduntur, rationi patent.

Sic proximae cujusque effectus causae indagantur, harum natura ex indole
collectorum, cognitorum et comparatorum phaenomenon indagata
perficitur, firmatur, et sensim ex horum aggregato consummatur tandem.

Quid speratis futurum, qui ad hanc normam sua exigit studia?

Nonne immutabilis et coeva erit haec scientia ipsi naturae humanae, ex
cujus sc. elicitur indole, in qua fundatur tantum?

Nonne certa erit, quae innixa iis, quae omnes pari agnoscunt evidentia,
castigatissima caute procedit fide?

Nonne definita satis et ipsis erit rebus utilis, quae certis, claris, et
sensibilibus corporis humani proprietatibus solum debet causae proximae,
quaeque nostro subjicitur imperio, inquisitionem accuratissimam, idque via,
qua erratum nunquam?

Lento crescet, fateor, et occulto adolescet augmento, quilibet tamen vel
minimus progressus gradus ad altiora firmus erit, et novi incrementi
immutabilis causa.

Hoc autem labore defunctum, adspirantemque ad metam jam videte in
ultima picturae parte adumbratum.

In ipsa nunc adyta se penetrat, in ipsa Æsculapii penetralia!

En Tabulas Hippocraticas, fidaque Grajorum, quae scrutatur, scripta!

Jam ex abundanti Medicorum Thesauro colligit quidquid sparsum haeret
mellis medicati.

Hic incisa, quorum notaverat morbos, ruspatur cadavera; illic in brutis arte
factas aegritudines observat; nunc omnia morborum effecta et remediorum
ipse experimento colligens; nunc eadem ex optimis Auctoribus addiscens;
tandem cuncta digerens, expendensque inter se componit, et his, quae

Theoria demonstravit, comparat, unde historiam denique curationemque morborum firmet.

En Vobis ultima manu absolutam consummati Medici imaginem!

Hanc Mechanicis egere auxiliis ut perficiatur, satis, ni fallit me animus, evictum.

Huic consimilem me reddere, ad hanc me componere studui, ut medicinam feci.

Ad hanc polire eorum, qui meae se committunt disciplinae, ingenium summa ope enixus sum, dum in Vestro hoc salutis fano ex Auctoritate vestra Musagetæ Illust. medicinam docui.

Eam, dum Dei munere spiro, ambitiose colere non desinam.

Non credulitate stulta, non stupore ignari vulgi, non verbosis strophis, sed clara demonstrationis fide Artem, cui nostra credimus capita, commendare affectabo.

Vos Optimi Juvenes, qui illi Scientiæ consecrastis pectora, a qua incolumitatem sperat salutis Humanum Genus, Vos Picturam. Medici contemplati primis miremini ab annis.

Ita Vos agite rem vestram, ut lineamentis, coloribusque hujus imaginis formosi, salutare hominibus audiat genii!

Nulla est, quæ pulchriora laborum præmia Cultoribus persolvit, quam Medica Sapientia.

Non alia est, quæ Mortalibus gratiores, magisve utiles vel necessarios reddere vos possit.

Excitemini o generosæ mentes! Excitemini pulchritudine Artis, cujus effectu beatus his in terris nemo carere poterit!

Nunquam rei difficultas calidum vestri animi retundat impetum!

Ardua est, fateor, quæ ad Panacææ ducit delubra, via.

Sed complanavit hanc improbus aliorum labor, superarunt praerupta, perrupere fortes, Vos alacres sequamini!

Hos habetis in hac Academia ad Medicinam Duces, qui ditiores longe Vobis explicant thesauros, quam Epidauriae olim columnae, Pergamenae tabulae, Cnidii parietes, vel folia largiebantur Coaca.

Habetis, qui secreta quaeque Matheseos arcana incredibili perspicui sermonis facilitate revelet, rebusque applicare Medicis praemonstret, Volderum.

Optimorum sane sententia natum ad haec sacra, Nostroque encomio longe majorem Virum!

Cujus disciplinae liberali infinitum me debere grata memoria et publice hic agnosco, et dum huic constabit menti sanitas ingenue semper Ego et candide meminero.

Horum ergo dum lego vestigia, si quid vobis adjumenti praestare posse censeor, praesto sum qui ita me geram, ut ex vestro meum me comparare commodum opere ipso testari possim.

Vobiscum Veterum placita, Recentiorum et propria, si quae sunt, observata undique indefesso labore colligere, ex his laudatae Mechanices arte doctrinam Medicam condere non desinam, quamdiu in hac versanti slatione, vires dederit Deus!

Agite ergo Commilitones Studiosi totus quod commendavit sermo, felici hujus anni Academici auspicio inchoare et perficere certatim tentemus opus!

Vestra frequentia incitatus docentis vigor id aget, ut, qui naturae facultate et eruditionis plurimis postponendum me sentio, sedulitate certe cedam nulli.

Laboris autem summum habebo pretium, si vestro applausu, Vobis meam profuisse diligentiam, orbi constet, si vestri in hoc Athenaeo studii felicitas claritate famae plures alliciat.

Hoc enim votum illud est, *Illustrissimi Curatores, Amplissimi Coss.*, cujus successu alacer, rerum Vestro auspicio, Vestra in Academia gestarum rationem Vobis reddere audebo.

Unum hoc dignum habebo, quo Genium Vestrum adorem, donarium.

Omni sic adulationis fuco deterso, sincero certe animi candore referre me putabo, quas Vestrae benignitati animus debet, gratias!

Docendi enim admotum muneri, duoque jam meritum stipendia, exploratum adeo, honorificis promissis et nova liberalitate nec opinantem excitastis denuo.

Ego, ex multis, quas in Vobis veneror, virtutibus, unam prae caeteris eximiam habendam esse a Sapientibus accepi, sinceram nempe Vestri favoris integritatem.

Summam dico, et Reip. literariae solam salutarem Virtutem, qua praemia meritis, non gratiae servire jubetis, neque ambitioni.

Quare benefacti pretium Vestra ex gravitate ponderans, vix mihi tempero, quin tanti testimonii gloria animosus, quo coepi pede, pergam alacrior!

Verbosae ergo pompae loco, qua gratiarum actio suspecta redditur et Sapientibus odiosa, pauca ego haec religiosus spondeo!

Vestram Dignitatem summo venerationis cultu et obsequii semper colam sedulus!

Diligens sic mea se acuet industria, ut Vestrum favorem plurimi me facere et legitimis ultra ambire artibus, demonstrem.

Id studebo, ut bene agendo benefici, quod de me tulistis, iudicii aequitatem Orbi ipse comprobem!

DIXI.

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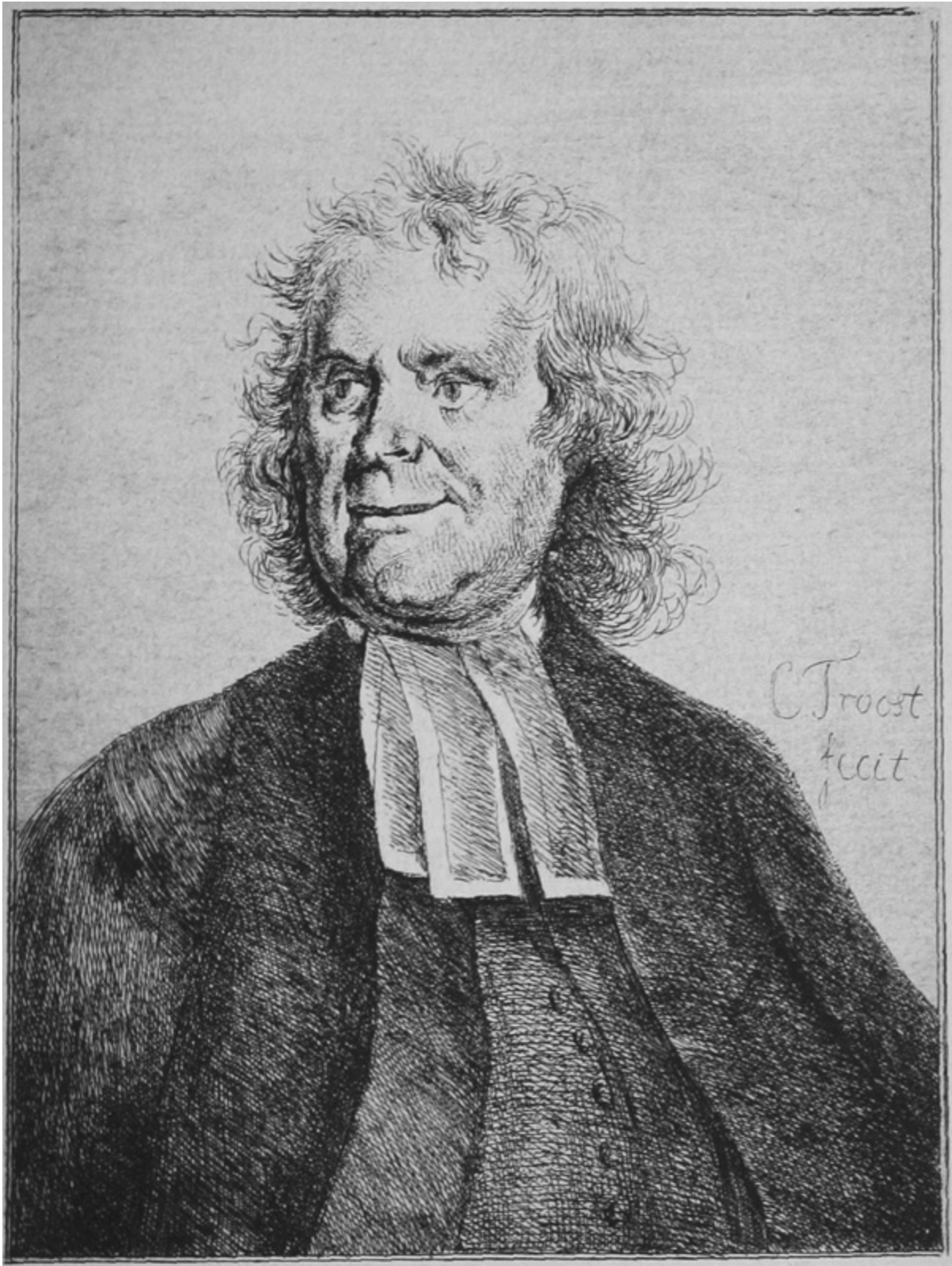
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